AnalystLab Africa – Week 3 Summary Report

SQL & Data Querying: Chinook Database and Sales Dataset

Overview

For this task, I used PostgreSQL to work with two datasets: the Chinook database (a music store) and a Sales dataset. I wrote SQL queries to filter, sort, group, join, and analyze the data, and I also looked at how to make queries run faster.

Core SQL Queries

I used SELECT, WHERE, and ORDER BY to filter and sort data. For example, I found all customers from Brazil and sorted them by city. I also used GROUP BY and HAVING together with COUNT, SUM, and AVG to summarize data, like counting how many customers are in each country and working out how much each customer spent in total and on average.

Advanced SQL Concepts

I used INNER JOIN to combine the customer and invoice tables, so I could see customer names next to their invoices instead of just ID numbers. I used a subquery to find customers who spent more than the average customer. I also used window functions like RANK() to rank invoices, both overall and separately for each customer.

Business Problem Solving

For the Chinook dataset, I found the best-selling tracks by revenue, how revenue changed year by year, and which countries bring in the most money and customers. For the Sales dataset, I found the best-performing product lines, how revenue changed month by month, and the top customers by how much they spent and how many orders they made.

Query Optimization

I used EXPLAIN ANALYZE to check how PostgreSQL runs a query. I noticed that searching by billing_city was slow at first because it had to check every row one by one. After I created an index on that column, the same search became much faster. This showed me how useful indexes are, especially for bigger datasets.

Key Insights

1. In the Chinook dataset, a few countries like the USA and Canada have the most customers, and a few tracks bring in a lot more money than the rest.

2. In the Sales dataset, some product lines make a lot more money than others, and a small number of customers make up a big part of total sales.

3. Adding indexes can make searches much faster, especially as the data gets bigger.

Conclusion

This task helped me understand how to use SQL not just to get the right answers, but also to think about speed and making queries run efficiently.